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NPS/001/015 – Technical Specification for Barbed Wire

1. Purpose

The purpose of this document is to detail the technical specification for barbed wire for use on the overhead line distribution networks of CE Electric UK.

This document supersedes the following documents, all copies of which should be destroyed.

Ref	Version	Title
DSS/004/002	1.0	Specification for Barbed Wire
NEDL/MS – 0/03		Specification for Barbed Wire for Overhead Lines
NPS/001/015	1.0	Technical Specification for Barbed Wire

2. Scope

This document applies to barbed wire for use on wood poles, towers and support stays on the overhead line networks of CE Electric UK.

3. Technical Requirements

3.1 General

Barbed wire shall be in accordance with Energy Networks Association Technical Specification (ENA TS) 43-90 Issue 4: 1993, manufactured in accordance with BS 4102: 1998 and galvanised to BS EN 10244 – 2: 2001.

3.2 Material Requirements

Barbed wire shall be 2 PLY with a 2.5mm diameter and have four pointed barbs. The barbs shall have a 2mm diameter and be 12.5mm in length. The spacing between the barbs shall not exceed 50mm.

The barbed wire will be long life and have a galvanised coating of 735g/m² of surface area as detailed in BS EN 10244 – 2 Table 1, Class A x 3^b.

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4. References

4.1 External Documentation

Reference	Title
ENA TS 43-90 Issue 4: 1993	Anti-climbing Devices and Safety Signs for HV Lines up to and including 400kV
BS 4102: 1998	Specification for steel wire general fencing purposes
BS EN 10244 – 2: 2001	Steel wire and wire products. Non-ferrous metallic coatings on steel wire. Zinc or zinc alloy coatings

4.2 Internal documentation

Reference	Title
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5. Definitions

Term	Definition
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6. Authority for issue

6.1 Author

I sign to confirm that I have completed and checked this document and I am satisfied with its content and submit it for approval and authorisation.

		Sign	Date
Steven Salkeld	Standards Engineer	Steven Salkeld	19/12/07

6.2 Policy Sponsor

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
Geoff Bolden	Purchasing Operations Manager	Geoff Bolden	20/12/07

6.3 Technical Assurance

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
Ged Hammel	Overhead Standards Manager	Ged Hammel	20/12/07

6.4 DBD Assurance

I sign to confirm that this document has been assured for issue on to the DBD system

		Sign	Date
Sean Johnson	DBD Administrator	Sean Johnson	11/01/08

6.5 Approval

Approval is given for the content of this document

		Sign	Date
Chris Holdsworth	Standards Manager	Chris Holdsworth	24/12/07

6.6 Authorisation

Authorisation is granted for publication of this document

		Sign	Date
Carol Morgan	Head of Procurement	Carol Morgan	27/12/07

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Appendix 1 – Self Certification Conformance Declaration

To be completed by suppliers.

Barbed wire shall comply with the latest issues of the relevant international and British Standards.

This check sheet identifies the clauses in BS 4102 and BS EN 10244 - 2 and the clauses of the aforementioned Standards relevant to barbed wire for use on current United Kingdom designs of wood pole and tower overhead lines.

The manufacturer shall declare conformance or otherwise, clause by clause, using the following levels of conformance declaration codes.

Conformance declaration codes

N/A = Clause is not applicable/ appropriate to the product

Cs1 = The product conforms fully with the requirements of this clause

Cs2 = The product conforms partially with the requirements of this clause

Cs3 = The product does not conform to the requirements of this clause

Cs4 = The product does not currently conform to the requirements of this clause, but the manufacturer proposes to modify and test the product in order to conform.

Manufacturer:

Product Reference:

Instructions for completion

- When Cs1 code is entered no remark is necessary
- When any other code is entered the reason for non-conformance shall be entered
- Prefix each remark with the relevant 'BS EN' or 'BS' as appropriate

Name:

Signature:

Date:

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BS 4102: 1998			BS EN 10244 – 2: 2001			Remarks
Clause / Sub-clause	Requirement	Conformance Code	Clause / Sub-clause	Requirement	Conformance Code	
3.0	Materials		4.0	Coating requirements (Table 1 - Class A x 3 ^b)		
4.0	Wire diameter and tolerances		5.0	Testing		
5.0	Coatings					
6.0	Tensile strength					
7.0	Inspection and testing					
8.0	Markings					